

Arguing serialism and gradualness

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1 How not to do HS

Japanese¹ ([jpn](#), Japonic)²

¹ [ɲi.hoŋ.gó.]

- (1) **V ~ Ø alternations in Sino-Japanese compounds** (Itô and Mester, 1996:15-17; Tateishi, 1990; Itô and Mester, 2015)

² Thanks to Yasu Sudo for the accents which reflect Tokyo Japanese

/gak(u)/	study, learning	/bet(u)/	different
[gákɰ-i]	academic degree	[bétɰ-i]	different opinion
[gakú-jú:]	school friend	[betsú-jákú]	different translation
[gakú-réki]	academic background	[bétɰ-ri]	separation
[gakú-nén]	school year	[betsú-nó:]	separate payment
[gakú-bátsú]	academic clique	[betsú-bín]	separate carrier
[gákɰ-ha]	sect	[bép-pái]	farewell cup/dinner
[gakú-sái]	scholastic ability	[bes-só:]	separate mail/shipment
[gakú-tjó:]	college president, rector	[bet-tákú]	detached villa
[gak-kó:]	school	[bek-kákú]	different style

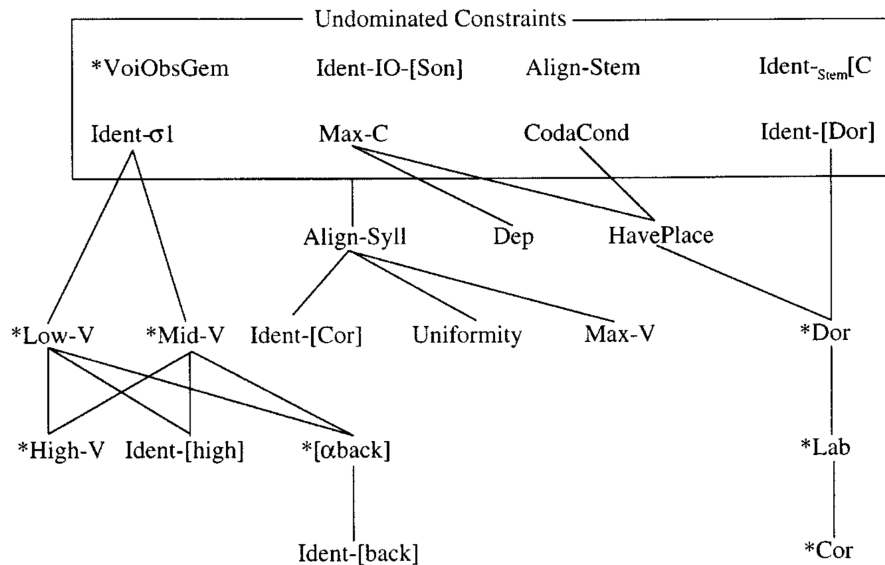
Distribution

- [Ø] / /k/ _ /k/
- [Ø] / /t/ _ [-voice]
- [ɰ] / elsewhere

Kurusu's (2000) parallel OT analysis

- ALIGN-SYLL: left edge of syllable = left edge of word
motivates general vowel syncope
- ALIGN-STEM: left edge of stem = left edge of syllable
blocks syncope with V-initial suffixes
- low ranked constraints produce [t] and [ɰ] if epenthetic
/ underspecified

(2) (63) Overall constraint ranking



(3) /hatu-bai/ → [.ha.tsu.bai.] (Kurusu, 2000:165)

/hatu-bai/	*VOIOBSGEM	CODACOND	IDENT(son)	IDENT/stem[C]	ALIGN-SYLL	IDENT(cor)	MAX(V)
a. [⟨.ha.tsu.bai.⟩]					3		
b. [⟨.hab.bai.⟩]	W 1				L 1	W 1	W 1
c. [⟨.hat.bai.⟩]		W 1			L 1		W 1
d. [⟨.ham.bai.⟩]			W 1		L 1	W 1	W 1
e. [⟨.hat.tai.⟩]				W 1	L 1		W 1

Adler and Zymet (2021:394): “The boundary-adjacent vowel deletes, and the resulting cluster undergoes gemination [...] But whenever this would produce a voiced geminate, the compound surfaces faithfully”

- The map *is not* the territory
the generalizations are the SRs and the mappings
if a grammar models them, it works.

Kurusu’s (2000) analysis does not work in HS

- this is not unusual
*TONELESSVOWEL_x motivates iterative spreading in OT
cannot in HS assuming spreading is gradual
bátaga ↛ bátágá in one step
and banning gapping contra Kimper (2012)

(4) This doesn't work (Adler and Zymet, 2021:395)

	*VOIBSGEM	CODACOND	ALIGN-SYLL	MAX(V)
a. /betu-bin/ → betu-bin ~ bet-bin		W	L	W
b. /betu-pai/ → bet-pai ~ betu-pai		L	W	L

- EVAL cannot look ahead to anticipate later steps
⇒ local (non-global) optima

Is this evidence of ...anything?

- not really
in the context of their paper, it supports parallelism
- $\exists \nrightarrow \forall$
there are a lot of analyses that do not work
here's one that works

Constraints

- *TVT:³ AOVFE vowel between voiceless obstruents
(or high vowels specifically, or add faithfulness)
- *RISINGSONORITY:⁴ AOVFE obstruent-sonorant coda-onset cluster
- *[K,P,T]{K,P,T}:⁵ AOVFE heterorganic cluster
- *[K]{K,P,T}: AOVFE dorsal-initial heterorganic cluster
- IDENT{K,P,T}
- IDENT{K}
- MAX(V):⁶ AOVFE deleted vowel
- DEP(V): AOVFE inserted vowel

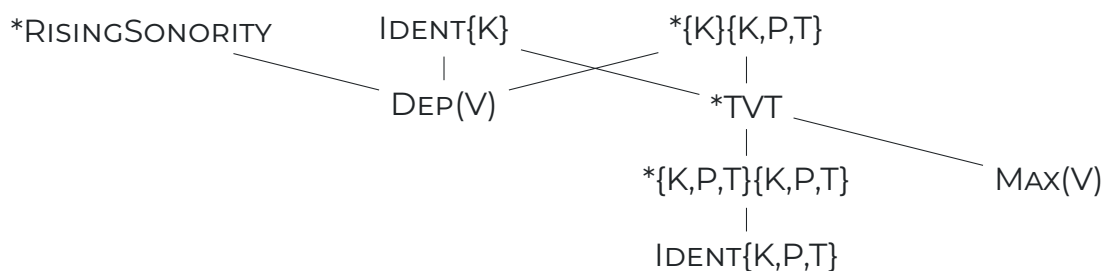
³ cf. Tsuchida (1997:§2.4.2)

⁴ see Gouskova (2004)

⁵ Under the non-autosegmental interpretation (de Lacy, 2006:183)

⁶ (McCarthy and Prince, 1994:342)

(5) **Hasse diagram illustrating the ranking**



omitting ALIGN-STEM which explains [u-i] for space

underlying status of [u-] is unclear

Operations: insert/delete u, change place

(6) /gaku-ko:/ → [.ga.kó:] (a-d); /gaku-ko:/ → [.gák.kó:] (c-d)

	*RISINGSONORITY	IDENT{K}	*{K}{K,P,T}	DEP(V)	*TVT	*{K,P,T}{K,P,T}	IDENT{K,P,T}	MAX(V)
/gaku-ko:/								
a. [.ga.ku.ko:]					W 1			L
☞ b. [.gák.ko:]								1
☞ c. [.gák.ko:]								
d. [.ga.ku.ko:]				W 1	W 1			

(7) /gak-nen/ → [.ga.kú.nén.] (a-e); /gaku-nen/ → [.ga.kú.nén.] (d-e)

	*RISINGSONORITY	IDENT{K}	*{K}{K,P,T}	DEP(V)	*TVT	*{K,P,T}{K,P,T}	IDENT{K,P,T}	MAX(V)
/gak-nen/								
a. [.gák.nén.]	W 1		W 1	L		W 1		
b. [.gat.nén.]	W 1	W 1		L			W 1	
☞ c. [.ga.ku.nén.]				1				
☞ d. [.ga.ku.nén.]								
e. [.gák.nén.]	W 1		W 1			W 1		W 1

(8) /gak-sai/ → [.ga.kú.sai.] (a-e); /gak-sai/ → [.ga.kú.sai.] (d-e)

	*RISINGSONORITY	IDENT{K}	*{K}{K,P,T}	DEP(V)	*TVT	*{K,P,T}{K,P,T}	IDENT{K,P,T}	MAX(V)
/gak-sai/								
a. [.gák.sai.]			W 1	L	L	W 1		
b. [.gas.sai.]		W 1		L	L		W 1	
☞ c. [.ga.ku.sai.]				1	1			
☞ d. [.ga.ku.sai.]					1			
e. [.gák.sai.]			W 1		L	W 1		W 1

- onset faithfulness rules out .gak.kai.

(9) /betu-taku/ → [.bet.tá.kú.] (a-d); /bet-taku/ → [.bet.tá.kú.] (c-d)

/betu-taku/	*RISINGSONORITY	IDENT{K}	*{K}{K,P,T}	DEP(V)	*TVT	*{K,P,T}{K,P,T}	IDENT{K,P,T}	MAX(V)
a. [.be.tu.ta.ku.]					W 1			L
☞ b. [.bet.ta.ku.]								1
☞ c. [.bet.ta.ku.]								
d. [.be.tu.ta.ku.]				W 1	W 1			

(10) /bet-no:/ → [.bé.ṭsú.nó:] (a-d); /betu-no:/ → [.bé.ṭsú.nó:] (c-d)

/bet-no:/	*RISINGSONORITY	IDENT{K}	*{K}{K,P,T}	DEP(V)	*TVT	*{K,P,T}{K,P,T}	IDENT{K,P,T}	MAX(V)
a. [.bet.no:]	W 1			L				
☞ b. [.be.tu.no:]				1				
☞ c. [.be.tu.no:]								
d. [.bet.no:]	W 1						W 1	

(11) /betu-kaku/ → [.bek.ká.kú.] (a-g); /bet-kaku/ → [.bek.ká.kú.] (c-g)

/betu-kaku/	*RISINGSONORITY	IDENT{K}	*{K}{K,P,T}	DEP(V)	*TVT	*{K,P,T}{K,P,T}	IDENT{K,P,T}	MAX(V)
a. [.be.tu.ka.ku.]					W 1	L		L
☞ b. [.bet.ka.ku.]						1		1
c. [.bet.ka.ku.]						W 1	L	
☞ d. [.bek.ka.ku.]							1	
e. [.be.tu.ka.ku.]	W 1			W 1			L	
☞ f. [.bek.ka.ku.]								
g. [.be.ku.ka.ku.]				W 1	W 1			

- tie-in to yesterday: anti-heterorganic constraints (à la AGREE(place)) cannot be replaced with CODA COND
- sandwich *TVT in ranking

2 Calculating typologies

Learn Python; Hammond (2020) is a great resource

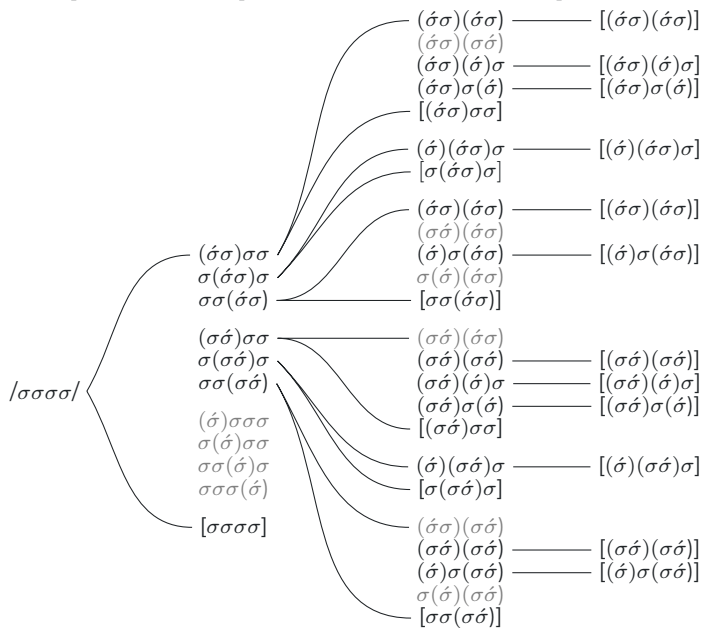
- Code repo including implementation of FRed (Brasoveanu and Prince, 2011)

<https://github.com/aphonologist/directionalHSfooting>

- there is existing software, but it can be inflexible

Discussion below from Lamont (2022)

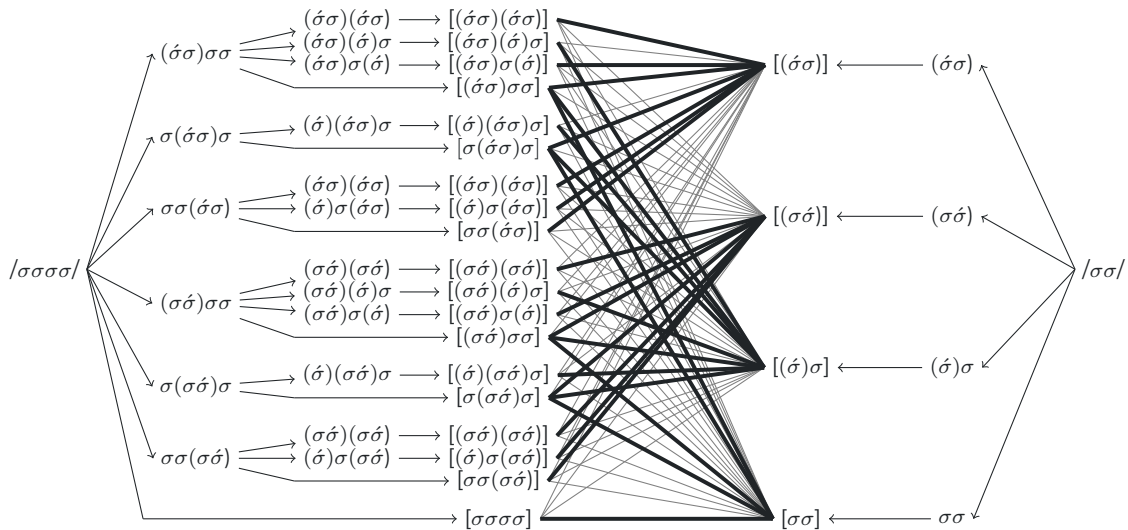
(12) Possible derivations from /σσσσ/; candidates in gray are not possible optima, and are not passed back into GEN



(13) Comparison of optimal and possible surface forms and derivations generated from inputs of various lengths

Input length	Optimal surface forms	Optimal derivations	Possible surface forms	Possible derivations
2	4	4	6	7
3	9	9	16	28
4	17	19	44	139
5	21	29	120	814
6	24	35	328	5,527
7	36	55	896	42,712
8	36	55	2,448	370,507
9	31	55	6,688	3,567,034

- (14) **Combining the derivations generated from four and two syllable strings; combinations with thick black lines have satisfiable ranking conditions, and those with thin grey lines require inconsistent rankings**



- (15) **Comparison of optimal and possible combinations of derivations for inputs of at least two syllables**

Input length	Optimal combinations	Total checks	Possible combinations
2	4	—	4
3	13	36	196
4	35	247	27,244
5	67	1,015	2.2×10^7
6	103	2,345	1.2×10^{11}
7	131	5,775	5.2×10^{15}
8	151	7,205	1.9×10^{21}
9	179	8,305	6.9×10^{27}

3 Autosegmental shift

Kibondei ([bou](#), Niger-Congo, Tanzania) – discussion from Lamont (2025)

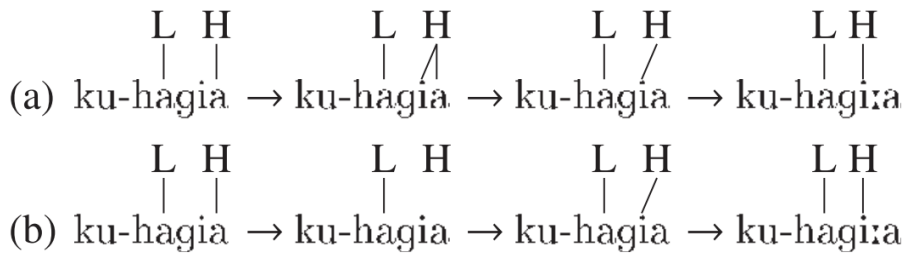
- (16) Phrase-final shift in Kibondei (Kisseberth and Cassimjee, 2006:84-85)

(a) $\begin{array}{cc} \text{L} & \text{H} \\ | & | \\ \text{ku-hagia} & \rightarrow \text{ku-hagia} \end{array}$ ‘to sweep’

(b) $\begin{array}{cc} \text{L} & \text{H} \\ | & | \\ \text{ku-hagia-na} & \rightarrow \text{ku-hagia-na} \end{array}$ ‘to sweep for one another’

one step (Gietz et al., 2023) or two (McCarthy, 2006, 2007)?

(17) Shift as linking+delinking (a) or delinking+linking (b)



Argument for two-step from typologies

- <https://github.com/aphonologist/hs-shift>

- GEN

delete features

delete/insert links

±single step shift

- CON

MAX(link), DEP(link), MAX(F), MAXLINKED(F)

*FLOAT, *LINK(F), *F], ALIGN-R

only without single step shift / only with single step shift
 / $\Rightarrow$ = spreading / $\rightarrow$ = shifting

(18) The typology as input-output mappings

	Pre-penult	Penult	Final	Floating
(1)	$\Rightarrow$ final	$\Rightarrow$ final	faithful	$\rightarrow$ final
(2)	$\Rightarrow$ final	$\Rightarrow$ final	faithful	delete
(3)	$\Rightarrow$ final	$\Rightarrow$ final	faithful	faithful
(4)	$\rightarrow$ final	$\Rightarrow$ final	faithful	$\rightarrow$ final
(5)	$\rightarrow$ final	$\rightarrow$ final	faithful	$\rightarrow$ final
(6)	$\rightarrow$ final	$\rightarrow$ final	faithful	delete
(7)	$\Rightarrow$ penult	faithful	$\rightarrow$ penult	$\rightarrow$ penult
(8)	$\Rightarrow$ penult	faithful	$\rightarrow$ penult	delete
(9)	$\Rightarrow$ penult	faithful	delink	faithful
(10)	$\Rightarrow$ penult	faithful	delete	$\rightarrow$ penult
(11)	$\Rightarrow$ penult	faithful	delete	delete
(12)	$\Rightarrow$ penult	faithful	faithful	$\rightarrow$ penult
(13)	$\Rightarrow$ penult	faithful	faithful	delete
(14)	$\Rightarrow$ penult	faithful	faithful	faithful
(15)	$\rightarrow$ penult	faithful	$\rightarrow$ penult	$\rightarrow$ penult
(16)	$\rightarrow$ penult	faithful	$\rightarrow$ penult	delete
(17)	$\rightarrow$ penult	faithful	delete	$\rightarrow$ penult
(18)	$\rightarrow$ penult	faithful	delete	delete
(19)	$\rightarrow$ penult	faithful	faithful	$\rightarrow$ penult
(20)	$\rightarrow$ penult	faithful	faithful	delete
(21)	delink	$\Rightarrow$ final	faithful	faithful

(22)	delink	delink	delink	faithful
(23)	delink	delink	faithful	faithful
(24)	delete	$\Rightarrow$ final	faithful	$\rightarrow$ final
(25)	delete	$\Rightarrow$ final	faithful	delete
(26)	delete	$\rightarrow$ final	faithful	$\rightarrow$ final
(27)	delete	$\rightarrow$ final	faithful	delete
(28)	delete	delete	faithful	delete
(29)	delete	delete	delete	delete
(30)	faithful	faithful	$\rightarrow$ penult	$\rightarrow$ penult
(31)	faithful	faithful	$\rightarrow$ penult	delete
(32)	faithful	faithful	delink	faithful
(33)	faithful	faithful	delete	$\rightarrow$ penult
(34)	faithful	faithful	delete	delete
(35)	faithful	faithful	faithful	$\rightarrow$ final
(36)	faithful	faithful	faithful	$\rightarrow$ penult
(37)	faithful	faithful	faithful	delete
(38)	faithful	faithful	faithful	faithful

weird: fewer languages with more operations

(19) The typology as sets of output forms

(i) Long span at end	$\emptyset_0 F_0$	1, 2, 3
(ii) Short span at end	$\emptyset_0 F_0^2$	4, 21, <u>24</u> , 25
(iii) Long span at penult	$\emptyset_0 F_0 \emptyset$	7, <u>8</u> , 9, 10, 11
(iv) Long span at penult or singleton at end	$\emptyset_0 F_0 \emptyset \cup \emptyset_0 F$	12, 13, 14
(v) Singleton anywhere	$\emptyset_0 F_0^1 \emptyset_0$	35, 36, 37, 38
(vi) Singleton non-finally	$\emptyset_0 F_0^1 \emptyset$	30, <u>31</u> , 32, 33 , 34
(vii) Singleton at penult or end	$\emptyset_0 F_0^1 \emptyset_0^1$	19, 20
(viii) Singleton at end	$\emptyset_0 F_0^1$	5, 6, 23, 26, 27, 28
(ix) Singleton at penult	$\emptyset_0 F_0^1 \emptyset$	15, <u>16</u> , 17, 18
(x) No features	$\emptyset_0$	22, 29

(vii) is only unique pattern; requires shift to be two steps

Wolaitta ([wal](#), Afro-Asiatic, Ethiopia; Amha, 1996)

(20) (39) *Final (i) and penultimate (ii) tone in Wolaitta* (Amha 1996: 115–116)

	H		H		
(a) (i)	zare	'lizard'	(ii)	zare	'relative'
	H		H		
(b) (i)	ʃooro	'neighbour'	(ii)	siire	'nose'
	H		H		
(c) (i)	mifira	'bride'	(ii)	namisa	'hunger'
	H		H		
(d) (i)	karaabe	'drum'	(ii)	malaata	'sign'

(21) (40) *Shift to penult in Wolaitta as linking+delinking (hypothetical)*

	H pataka	MAX(H)	*FLOAT	*H] _φ	ALIGN-R	*LINK(H)	MAX(link)	DEP(link)
	a. H pataka				W 2	L 1		L
→	b. H pataka				1	2		1
	c. H pataka		W 1		L	L	W 1	L
	d. H pataka	W 1			L	L	W 1	L
→	e. H pataka				1	W 2	L	
	f. H pataka			W 1	L	W 3	L	W 1
→	g. H pataka				1	1	1	
→	h. H pataka				1	1		

(22) (41) *Final high tones surface faithfully*

	H mifira	MAX(H)	*FLOAT	*H] _φ	*LINK(H)	MAX(link)
→ a.	H mifira			1	1	
b.	H mifira		W 1	L	L	W 1
c.	H mifira	W 1		L	L	W 1

(23) (42) *Final high tones cannot surface faithfully with a single-step shift operation*

H mi ira	*H]_φ	ALIGN-R	DEP(link)	MAX(link)
☹ a. H mi ira	W 1	L	L	L
→ b. H mi ira		1	1	1

typology in Gietz et al. (2023) does not allow floating

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